

Advanced (Habanero)

- Q1.** What is the graph of $y = |x|$?
(A) A straight line
(B) A parabola
(C) A V-shaped graph
(D) A circle
(E) An ellipse
- Q2.** What is the vertex of $y = |x - 2|$?
(A) (0, 0)
(B) (0, 2)
(C) (2, 0)
(D) (2, 2)
(E) (0, -2)
- Q3.** What is the axis of symmetry of $y = |x + 1|$?
(A) $x = 0$
(B) $x = 1$
(C) $x = -1$
(D) $y = 0$
(E) $y = 1$
- Q4.** What is the direction of opening of $y = |x|$?
(A) Up
(B) Down
(C) Left
(D) Right
(E) None
- Q5.** What is the graph of $y = 2|x|$?
(A) A wider V-shaped graph
(B) A narrower V-shaped graph
(C) A straight line
(D) A parabola
(E) A circle
- Q6.** What is the effect of the transformation $y = |x| + 3$?
(A) Shifts the graph down by 3 units
(B) Shifts the graph up by 3 units
(C) Shifts the graph left by 3 units
(D) Shifts the graph right by 3 units
(E) No effect
- Q7.** What is the graph of $y = |x - 2| - 1$?
(A) A V-shaped graph shifted down by 1 unit
(B) A V-shaped graph shifted up by 1 unit
(C) A V-shaped graph shifted left by 2 units
(D) A V-shaped graph shifted right by 2 units
(E) A straight line
- Q8.** What is the equation of the graph that is a reflection of $y = |x|$ across the x-axis?
(A) $y = -|x|$
(B) $y = |x|$
(C) $y = -x$
(D) $y = x$
(E) $y = 0$

Open Ended Questions (FRQ)

- Q9.** Graph the function $y = |x + 2| - 3$ and identify its vertex, axis of symmetry, and direction of opening.
- Q10.** Compare and contrast the graphs of $y = |x|$ and $y = |x| + 2$. How do the transformations affect the vertex, axis of symmetry, and direction of opening?

Chef's Tips

- Tip 1: Use graph paper to help students visualize the graphs.
- Tip 2: Emphasize the importance of identifying the vertex, axis of symmetry, and direction of opening.
- Tip 3: Encourage students to use transformations to graph more complex absolute value functions.